

Economic Solutions for a Warming World: Correcting the Negative Externalities of Climate Change

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I Introduction

In January 2025, a new global record was set. The National Oceanic and Atmospheric Administration (NOAA) found that the average global surface temperature was 1.33°C or 2.39°F. This was the warmest temperature in January ever recorded since records began in 1850 (NOAA 2025). This wasn't the first record set in recent memory, the previous record was set in 2024 (NOAA 2025). Increasingly warmer temperatures like these reflect the existence of climate change, the long-term shift in temperature and weather patterns (NASA 2024). In fact, in the last 50 years, the global temperature has increased faster than at any point in the past 2,000 years (Jay et al. 2023). The consequences of climate change are drastic and will alter several aspects of everyday life. As time goes on, if we do not do anything to stop it, the effects of climate change will only worsen (Jay et al. 2023). In this paper, I will review what economics has to offer us as a solution. Specifically, I will discuss the negative externalities of climate change, economic theories that can help correct them, and how we apply those theories to the real world.

II The Externalities of Climate Change

What is an externality? Gruber (2022) provides an overview of externalities and explains why climate change is an environmental one. Gruber defines an externality as:

Whenever the actions of one party make another party worse or better off, yet the first party neither bears the costs nor receives the benefits of doing so.

Gruber states that an externality is negative when it makes another party worse off. Climate change clearly fits the definition of a negative externality; when firms produce greenhouse gasses, it has negative effects on others worldwide. Specifically, climate change is found to be a negative production externality, since the production of greenhouse gasses causes negative effects on others (who are not compensated at all by emitting firms). Gruber points

out that negative production externalities are market failures, and result in overproduction of greenhouse gasses. Further, the production of greenhouse gasses results in large social marginal costs and dead-weight loss, which leads to welfare loss.

What are some of the potential impacts of climate change? The U.S. Global Change Research Program (USGCRP) released their Fifth National Climate Assessment (NCA5) where they detail potential impacts to the U.S. economy. In the assessment, [Hsiang et al. \(2023\)](#) show that the increased frequency and intensity of natural disasters caused by climate change has resulted in losses in crops, infrastructure, and worker productivity. Further, climate change has reduced housing prices, student learning, and GDP growth. These damages are projected to increase as climate change worsens, and these impacts will not be felt evenly across the nation. Turning to environmental and human impacts, [Jay et al. \(2023\)](#) found that climate change is increasing the intensity and occurrence of drought, floods, hurricanes, extreme heat, sea level rise, and wildfire. Further, the authors found that these extreme events damage infrastructure, reduce human health, reduce water supplies, threaten food security, threaten ecosystems, and make agriculture harder. Much like the economic impacts, these effects will worsen as climate change goes on.

III Economic Theory

What does economic theory say we can do about negative externalities? In this section I will look at some potential solutions from the private sector and the public sector. Then, I will review the benefits and drawbacks of each policy and show what the current literature views as the best tool to correct the negative externalities of climate change.

a. Private Sector Solutions

[Gruber \(2022\)](#) suggests the Coase Theorem as a solution to negative externalities. The Coase Theorem states that when there are defined property rights and costless bargaining, then private markets can internalize the externality, and reach the socially optimal outcome.

However, as Gruber notes, this is impossible with global externalities. He outlines three issues with the Coasian solution. First is the free rider problem; when investment in a solution has personal costs but common benefit, individuals will under-invest in the solution. Second, there may be issues in assigning who is responsible for the externality. Third, it can be very hard to negotiate when there are many parties involved. If we tried a Coasian solution in the U.S. to try to get the thousands of greenhouse gas emitting firms to reduce emissions, we would certainly run into these roadblocks. Therefore, to combat climate change, the Gruber found that the government must intervene.

b. Public Sector Solutions

What might a government intervention look like? [Gruber \(2022\)](#) suggests corrective taxation. Corrective taxation is when the government taxes firms for each good they produce. In the context of climate change, this is called a carbon tax. For example, the government can tax polluting firms for each ton of carbon dioxide that they produce when making a good. Also, a tax can be placed on fossil fuels for how much greenhouse gasses are emitted when they are burned. The tax is set equal to the marginal damage, which causes a firm's private marginal cost curve to be raised to the level of the social marginal cost curve. This results in a socially optimal quantity of carbon dioxide and eliminates the dead-weight loss. Note that this achieves the same outcome as the private market solution but is not subject to the same barriers.

What does the economic literature say about implementing a carbon tax? [Timilsina \(2022\)](#) provides an overview of the history and current state of the literature on carbon taxes. Environmental benefits of a carbon tax include not only the reduction in carbon dioxide in the atmosphere, but also other pollutants like particulate matter, sulfur, and nitrogen. Timilsina showed results from [Woollacott \(2018\)](#) that estimate a carbon tax in the U.S. that increases 5 percent annually would net \$62-\$71 billion in health benefits and prevent between 8,000-

19,000 deaths. Potential benefits would be larger in areas that suffer from higher levels of pollution. Fiscal benefits include more revenue for governments, and a broader tax base that will have a tougher time evading the tax. Subsequently, the introduction of a carbon tax could help the governments of nations who suffer from tax evasion problems.

What are the drawbacks of a carbon tax? [Timilsina \(2022\)](#) highlighted 3 concerns: aggregate economic growth, distributional effects, and competitiveness of domestic firms. One of the main concerns with carbon taxes is that they can hurt economic growth. Indeed, earlier studies found that this was the case, barring any consideration of positive effects from the reduced effects of climate change. However, recent literature suggests that this might not be true. Timilsina found consensus in the literature that the impact on the economy can be determined by how the government uses the tax revenue. Two common approaches are considered: giving the revenue to households in lump sum transfers and using the revenue for investment and tax cuts to boost economic output. There is significant support for lump sum transfers. The Economist's Statement on Carbon Dividends, signed by thousands of economists—including 28 Nobel Laureates, 4 former Chairs of the Federal reserve, and 15 former Chairs of the Council of Economic Advisers—suggests that revenue should be redistributed this way. The literature finds that lump sum transfers will reduce economic impact, but it will still have negative consequences. Timilsina cites [Mahmood and Marpaung \(2014\)](#) who found that the reduction in GDP is lessened to 1.5 percent with lump sum transfers versus a reduction in GDP of 2.3 percent without any kind of redistribution. The second most commonly cited approach is to use the tax revenue to for investment in order to raise economic output. Further, a carbon tax that increases over time can benefit growth. Combining these two, [Ross \(2018\)](#) found that a carbon tax that increases by 5 percent each year along with revenue being used to cut capital taxes results in an increase in GDP. The economic effects of these revenue distributions should be considered based on how a nation's economy is structured. Economies that rely more on carbon intensive energy and

cannot easily switch to low carbon fuels will see a greater loss in GDP compared to nations who aren't.

There are also concerns on the distributional effects of a carbon tax. The literature finds that the tax could be regressive. Carbon taxes will cause energy prices to increase, and since low income households have a higher share of energy costs relative to high income households, they will have a more expensive tax burden. One way to address this is to implement the lump sum transfer idea mentioned above. Timilsina found a consensus among economists that a lump sum transfer would negate or even make the tax progressive (resulting in high income households having the bigger tax burden). [Fremstad and Paul \(2019\)](#) found that this would result in an increase in income for 98 percent of American earners in the lowest decile. Further, [Williams et al. \(2015\)](#) found that lump sum transfers would improve welfare in 3 of the 5 income quartiles. The opposite effect is found if revenue is used to cut capital taxes; it would make carbon taxes more regressive. A final potential drawback of a carbon tax would be its effect on the competitiveness of domestic firms. Domestic firms who export a large amount of goods may be less competitive versus foreign firms who are not subject to a carbon tax. Timilsina found that the literature showed ample evidence of this, along with possible emissions leakages; when firms move production to countries without a carbon tax or where it is less intensive. To address this concern, Timilsina lists several solutions. The most common of which is a border tax adjustment (BTA) or border carbon adjustment (BCA). A BTA is a tariff on imports from other countries who do not have any policy to address climate change. Effects of a BTA vary across the literature, ranging from little impact on countries like China, but large consequences for developing nations. Other revenue measures include payroll and corporate income tax cuts and tax credits.

Other possible tools to reduce the emission of greenhouse gasses is an emissions trading system (ETS) and quantity regulation. [Gruber \(2022\)](#) suggests that regulation of the quantity of a pollutant is hard to implement effectively in practice. In order to mandate the

correct reduction in pollutants, the government would need to know each firm's marginal cost curve. In the case of climate change, this would be extremely difficult considering the sheer number of firms that emit greenhouse gasses. To get around this massive information issue, Gruber suggests the government could issue permits that allow a defined amount of pollution. Firms could then trade these permits on the market, allowing for firms with a higher marginal cost of abating pollution to pollute slightly more than regulation while firms with a lower marginal cost pollute slightly less. In the end, the externality is internalized at a lower cost to society than if there was a uniform reduction across all firms. Note that this is also viewed as a pricing tool, especially if the emissions permits are auctioned by the government.

What are the benefits and drawbacks of emissions trading? [Timilsina \(2022\)](#) found that an ETS and carbon taxes share many of the same impacts, if done correctly. Timilsina suggested that it will be no different than a carbon tax in both economic and environmental impact. This could be the case if emission permits are auctioned off by the government for revenue. Further, he cited several studies that found the competitiveness and burden effects will also be the same if both policy instruments are designed the same. However, it is suggested that an ETS can be hard to enforce. The government must monitor the emissions market, which can be very costly.

Which policy instrument is best suited to address climate change? [Gruber \(2022\)](#) lays out a situation where the government is trying to correct the negative externalities of climate change. However, there is uncertainty over the marginal cost of pollution and the government estimates a marginal cost curve lower than what it is in reality. Gruber found that the potential dead-weight loss from carbon taxes is very small. This is because the social marginal benefit curve for climate change is flat; the benefit from the abatement of an additional unit of greenhouse gas is small. Conversely, if the government implements a quantity regulation, the costs of reduction largely outweigh the benefits, and there is large dead-weight loss. When

trying to curb emissions, regulation will ensure that the exact correct amount of greenhouse gas abatement is achieved, but it will come at a large cost. A carbon tax will result in a lower amount of abatement, but at a much smaller cost. Gruber argues the lower abatement from a carbon tax is acceptable since the social benefits of an additional unit of abatement are small. [Timilsina \(2022\)](#) also compares the two types of tools. As mentioned by Gruber, the literature agrees that the choice of each policy instrument depends on the marginal benefit and marginal cost curves. In the context of climate change, Timilsina found a consensus in the literature that carbon taxes are preferred to quantity regulations. Specifically, pricing instruments have been found to be more efficient, cheaper, and less regressive. Other strengths of carbon taxes compared to regulation highlighted by Timilsina include that they cover every source of emission, incentivize innovation of low carbon technology, and prevent emissions leakages.

Gruber and Timilsina showed that carbon taxes are preferred to quantity regulation. If the solution is pricing tools, which one is best? In his analysis of the literature, [Timilsina \(2022\)](#) found a few interesting points on this. [Hoel and Karp \(2002\)](#) find that carbon taxes are preferred when there is a high discount rate and stock decay rate. Further, [Newell and Pizer \(2003\)](#) found that if the stock of a pollutant is large relative to its annual production, as is with greenhouse gasses, a corrective tax is preferred. Newell and Pizer also found that if substantial reduction of pollution is needed in the short run, an ETS is preferred. Turning to implementation, [Timilsina \(2018\)](#) suggested that a carbon tax will be easier to levy than an ETS if it is uniform across all sectors of an economy. A carbon tax will require less enforcement monitoring than an ETS, which may reduce potential costs from legal challenges. However, Timilsina noted that the perception of an ETS may be better than a carbon tax, since taxes are usually viewed as burdens and an ETS could be seen as a market opportunity. However, as mentioned above, if the instruments are implemented correctly, both will have the same economic and environmental impacts.

IV Implementation

As discussed numerous times above, implementation plays a crucial role in the cost and effectiveness of both carbon taxes and emissions trading systems. If done correctly, the two policy instruments will have the same effects, if not, there can be economic and environmental harm. In this section, I will review the successes and failures of real world implementations of these policy tools.

In 2005, the European Union launched an ETS called the EU Emissions Trading System (EU ETS). Doubts have been voiced over its effectiveness, however. [Bayer and Aklin \(2020\)](#) address these doubts and study whether or not the EU ETS has been effective in reducing emissions. The authors found that the largest concern was the price of emissions permits was too low compared to the social cost of carbon, which may threaten the EU ETS's ability to reduce carbon emissions. Comparing observed reductions to a scenario where there was no EU ETS, the authors found that the system reduced 1.2 billion tons of carbon dioxide from 2008 to 2016, nearly half of what the EU's target was in the Kyoto agreement. Further, in the absence of the EU ETS, it was found that the EU would emit 11.5 percent less carbon dioxide than without the system. The EU is, however, working to address issues with its system, such as delaying the auction of 900 million emissions permits and the introduction of a Market Stability Reserve that acts like a central bank. The authors do note there is skepticism by academics on whether this will work. Bayer and Aklin note that for a ETS to work, there must be strong political commitment to reducing emissions, and scarcity in carbon markets when needed.

[Timilsina \(2022\)](#) found that 30 economies worldwide have implemented a carbon tax, though there are several limitations. Timilsina reported [World Bank \(2020\)](#) data that shows 75 percent of carbon taxes are too low to reach goals set by the Paris Climate Agreement. Timilsina also uses the data to show that some carbon taxes are as high as \$199 USD in

Sweden, and as low as \$1 USD in Poland. Contrary to the theory discussed above about uniform tax rates across sectors, many economies have granted exceptions to certain industries. For example, Mexico's carbon tax covers all sectors but natural gas. Finally, revenue recycling programs vary across the world. Sweden uses their revenue to offset personal and corporate income taxes, while Chile uses theirs for education and social programs. Timilsina notes that the popularity of carbon taxes among policy makers has increased in recent years, despite past reluctance based on the political challenges of implementation.

Climate change has been addressed at the international level as well. The first major attempt was the Kyoto Protocol in 1997. Gruber (2022) provides an overview of how the Protocol worked. 38 nations agreed that by 2010, they would reduce their greenhouse gas emissions to 5% below 1990 levels. The Kyoto Protocol used an emissions trading system on an international scale. Countries could buy and sell permits to allow more emission, as long as the total goal was met. The United States, however, was the only country that agreed, but did not ratify the protocol. Gruber cited findings by Nordhaus and Boyer (2000) that found the targets would cost the U.S. over \$1 trillion. This led Nordhaus and Boyer to conclude that the U.S. bore a much higher cost relative to the rest of the world. Gruber notes that the reluctance of ratification ignores the fact that the U.S. could've bought emissions permits to help curb the expensive cost.

Did the Kyoto Protocol work? Maamoun (2019) addresses this question directly. Maamoun compares the observed emissions of countries who ratified the protocol to a scenario where they did not ratify. It was found that the ratifying countries reduced emissions by 7% compared to the counterfactual. Further, reductions were still found to be achieved even with the U.S. withdrawal, which led Maamoun to suggest that the participation of some major countries will not render climate agreements useless. It should be mentioned, however, that global emissions are still rising. In fact, the average carbon dioxide level in the atmosphere reached 419 parts per million (ppm) in 2023, after decades of consistent increases (NOAA

2024). In fact, 2023 marked the twelfth year in a row carbon dioxide in the atmosphere grew by more than 2 ppm (NOAA 2024). This unprecedented increase carbon dioxide levels clearly stands in the face of Kyoto Protocol's effectiveness. Indeed, Maamoun (2019) suggests that the protocol was a "successful first step". Further international agreements will be needed to reduce greenhouse gas emissions.

V Conclusion

Climate change presents a serious danger to Planet Earth. As time goes on, new records are set that underscores the danger we face. To combat this, policy makers can implement pricing tools or directly regulate the quantity of emissions. The clear consensus among economists is that pricing tools are the best method to reduce emissions. The two most common pricing tools are carbon taxes and emissions trading systems. These tools have proved very effective in "internalizing" the externalities created by climate change, while quantity regulation is found to cause large dead-weight losses. Considering the costs, benefits, and comparisons of the two pricing instruments, they are relatively the same if implemented correctly. However, it may be easier to implement a carbon tax. On the other hand, an emissions trading system may be an "easier sell" to the public. Policy makers should consider their political situation and makeup of their economy before they choose what policy instrument to use.

Turning to real world implementation, there have been mixed results. The European Union's Emissions Trading System has been shown to reduce emissions, and policy makers are working to fix issues with their system. Carbon taxes have had a wide range of variation in implementation. The exemption of certain sectors undermines the effectiveness of carbon taxes. Further, some countries may not be using the revenue in the most efficient way; this could be due to political pressures, however. International agreements such as the Kyoto Protocol have proved somewhat successful, though carbon emissions have continued to grow

quickly over the past decades. Further, these agreements are prone to withdrawals, like that of the U.S. in the Kyoto and Paris agreements. This may not render them useless, however, as suggested by [Maamoun \(2019\)](#). Widespread, correct use of carbon taxes and emission trading systems, along with international climate agreements, will be needed if we are to fully correct the negative externalities of climate change.

Works Cited

- Bayer, Patrick, and Michaël Aklin. “The European Union Emissions Trading System Reduced CO₂ Emissions Despite Low Prices.” *Proceedings of the National Academy of Sciences*, vol. 117, no. 16, 6 Apr. 2020, pp. 8804–8812, <https://doi.org/10.1073/pnas.1918128117>.
- “Economists’ Statement on Carbon Dividends Organized by the Climate Leadership Council.” *Economists’ Statement on Carbon Dividends Organized by the Climate Leadership Council*, The Climate Leadership Council, Jan. 2019, www.econstatement.org/.
- Fremstad, Anders, and Mark Paul. “The impact of a carbon tax on inequality.” *Ecological Economics*, vol. 163, Sept. 2019, pp. 88–97, <https://doi.org/10.1016/j.ecolecon.2019.04.016>.
- Gruber, Jonathan. *Public Finance and Public Policy*. Available from: VitalSource Bookshelf, (7th Edition). Macmillan Higher Education, 2022.
- Hanley, Nick, et al. *Introduction to Environmental Economics*. Available from: VitalSource Bookshelf, (3rd Edition). Oxford University Press Academic UK, 2019.
- Hoel, Michael, and Larry Karp. “Taxes versus quotas for a stock pollutant.” *Resource and Energy Economics*, vol. 24, no. 4, Nov. 2002, pp. 367–384, [https://doi.org/10.1016/s0928-7655\(02\)00014-3](https://doi.org/10.1016/s0928-7655(02)00014-3).
- Hsiang, S., et al. 2023: Ch. 19. “Economics”. In: *Fifth National Climate Assessment*. Crimmins, A.R., C.W. Avery, D.R. Easterling, K.E. Kunkel, B.C. Stewart, and T.K. Maycock, Eds. U.S. Global Change Research Program, Washington, DC, USA. <https://doi.org/10.7930/NCA5.2023.CH19>
- Jay, A.K., et al. 2023: Ch. 1. Overview: Understanding risks, impacts, and responses. In: *Fifth National Climate Assessment*. Crimmins, A.R., C.W. Avery, D.R. Easterling, K.E. Kunkel, B.C. Stewart, and T.K. Maycock, Eds. U.S. Global Change Research Program, Washington, DC, USA. <https://doi.org/10.7930/NCA5.2023.CH1>
- Maamoun, Nada. “The Kyoto Protocol: Empirical Evidence of a Hidden Success.” *Journal of Environmental Economics and Management*, vol. 95, May 2019, pp. 227–256, <https://doi.org/10.1016/j.jeem.2019.04.001>.
- Mahmood, Arshad, and Charles O.P. Marpaung. “Carbon Pricing and Energy Efficiency Improvement – Why to Miss the Interaction for Developing Economies? An Illustrative CGE Based Application to the Pakistan Case.” *Energy Policy*, vol. 67, Apr. 2014, pp. 87–103, <https://doi.org/10.1016/j.enpol.2013.09.072>.
- Newell, Richard G., and William A. Pizer. “Regulating Stock Externalities Under Uncertainty.” *Journal of Environmental Economics and Management*, vol. 45, no. 2, Mar. 2003, pp. 416–432, [https://doi.org/10.1016/s0095-0696\(02\)00016-5](https://doi.org/10.1016/s0095-0696(02)00016-5).

- “January Temperature Marks New Global Milestone”, National Oceanic and Atmospheric Administration, *National Oceanic and Atmospheric Administration*, 12 Feb. 2025, <https://www.noaa.gov/news/january-temperature-marks-new-global-milestone>.
- “No Sign of Greenhouse Gases Increases Slowing in 2023. NOAA Research”, *National Oceanic and Atmospheric Administration*, 5 Apr. 2024, research.noaa.gov/no-sign-of-greenhouse-gases-increases-slowng-in-2023.
- Nordhaus, William D., and Joseph Boyer. “Warming the World: Economic Models of Global Warming.” *MIT Press*, 18 Aug. 2000, <https://doi.org/10.7551/mitpress/7158.001.0001>.
- Ross, Martin T. “Regional implications of national carbon taxes.” *Climate Change Economics*, vol. 09, no. 01, Feb. 2018, p. 1840008, <https://doi.org/10.1142/s2010007818400080>.
- Timilsina, Govinda R. “Carbon Taxes.” *Journal of Economic Literature*, vol. 60, no. 4, 1 Dec. 2022, pp. 1456–1502, <https://doi.org/10.1257/jel.20211560>.
- Timilsina, G. R. “Carbon Pricing for Climate Change Mitigation.” *CABI Reviews*, 20 June 2018, pp. 1–13, <https://doi.org/10.1079/pavsnmr201813008>.
- “What Is Climate Change? - A Science.” Edited by Kalina Velev, NASA, National Aeronautics and Space Administration, 21 Oct. 2024, science.nasa.gov/climate-change/what-is-climate-change/.
- Williams, Robertson C., et al. “The Initial Incidence of a Carbon Tax Across Income Groups.” *National Tax Journal*, vol. 68, no. 1, 1 Mar. 2015, pp. 195–213, <https://doi.org/10.17310/ntj.2015.1.09>.
- Woollacott, Jared. “The economic costs and co-benefits of Carbon Taxation: A General Equilibrium Assessment.” *Climate Change Economics*, vol. 09, no. 01, Feb. 2018, p. 1840006, <https://doi.org/10.1142/s2010007818400067>.
- World Bank. *State and Trends of Carbon Pricing 2020*, 26 May 2020, <https://doi.org/10.1596/978-1-4648-1586-7>.